

**QUESTION ONE** (15 marks) Use a separate writing booklet.**Marks**

(a) Find  $\int_0^4 \frac{1}{\sqrt{2x+1}} dx$ . **2**

(b) Find  $\int \tan^3 x \sec^2 x dx$ . **2**

(c) Find  $\int \frac{x}{x^2 - 4x + 8} dx$ . **3**

(d) (i) Find the values of  $A$  and  $B$  such that  $\frac{3x^2 - 10}{x^2 - 4x + 4} = 3 + \frac{A}{x - 2} + \frac{B}{(x - 2)^2}$ . **2**

(ii) Find  $\int \frac{3x^2 - 10}{x^2 - 4x + 4} dx$ . **2**

(e) Use integration by parts twice, to show that  $\int_1^e \sin(\ln x) dx = \frac{e}{2}(\sin 1 - \cos 1) + \frac{1}{2}$ . **4**

**QUESTION TWO** (15 marks) Use a separate writing booklet.**Marks**

(a) Simplify  $|\cos \theta + i \sin \theta|$ . **1**

(b) Express  $\frac{i^5(1-i)}{2+i}$  in the form  $a + ib$  where  $a$  and  $b$  are rational. **2**

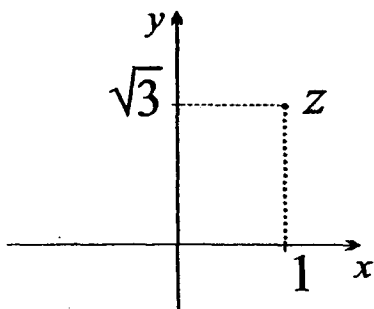
(c) By drawing a diagram, or otherwise, find the solutions of  $z^5 = -1$ . **2**

(d) Graph the region in the Argand diagram which simultaneously satisfies **3**

$$1 \leq |z - i| \leq 2 \text{ and } \operatorname{Im} z \geq 0.$$

(e) Find the complex number  $\phi$  if  $1 + i$  is a root of the equation  $z^2 + \phi z - i = 0$ . **2**

(f)



Suppose that  $z = 1 + \sqrt{3}i$  and  $\omega = (\operatorname{cis} \alpha)z$  where  $-\pi < \alpha \leq \pi$ .

(i) Find the argument of  $z$ . **1**

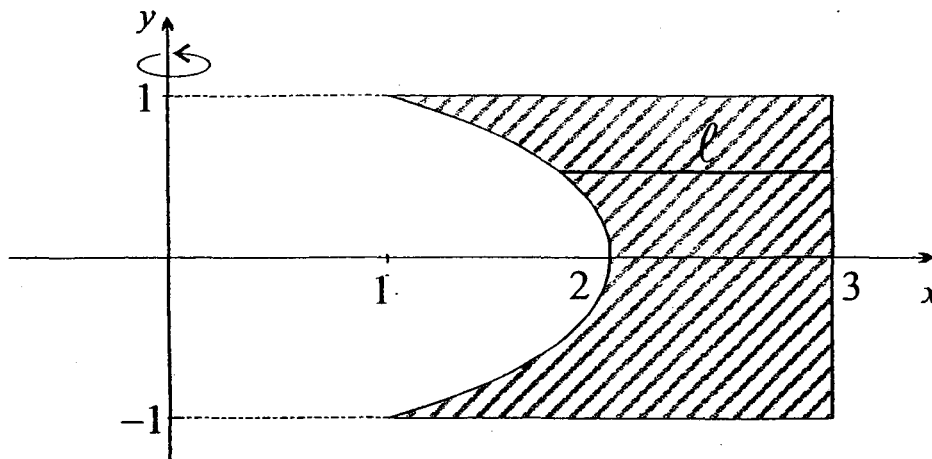
(ii) Find the value of  $\alpha$  if  $\omega$  is purely imaginary and  $\operatorname{Im}(\omega) > 0$ . **2**

(iii) Find the value of  $\arg(z + \omega)$  if  $\omega$  is purely imaginary and  $\operatorname{Im}(\omega) > 0$ . **2**

Exam continues next page ...

**QUESTION THREE** (15 marks) Use a separate writing booklet.**Marks**

(a)



The diagram above shows the region bounded by the curve  $x = 2 - y^2$  and the lines  $x = 3$ ,  $y = 1$  and  $y = -1$ . This region is rotated about the  $y$ -axis to form a solid. The interval  $\ell$  at height  $y$  sweeps out an annulus.

- (i) Show that the annulus at height  $y$  has area equal to

**2**

$$\pi(5 + 4y^2 - y^4).$$

- (ii) Find the volume of the solid.

**2**

- (b) Consider the function  $f(x) = \frac{1}{1+x^3}$ .

- (i) Show that there is a horizontal point of inflexion at  $x = 0$ .

**2**

- (ii) Find the vertical asymptote and the horizontal asymptote.

**2**

- (iii) Sketch  $y = f(x)$  showing the features from parts (a) and (b) and the  $y$ -intercept.

**2**

- (iv) On a separate diagram sketch  $y = |f(x)|$ .

**1**

- (v) On a separate diagram sketch  $y^2 = f(x)$ .

**2**

- (vi) On a separate diagram sketch  $y = e^{f(x)}$ .

**2**

Exam continues overleaf ...

**QUESTION FOUR** (15 marks) Use a separate writing booklet.**Marks**

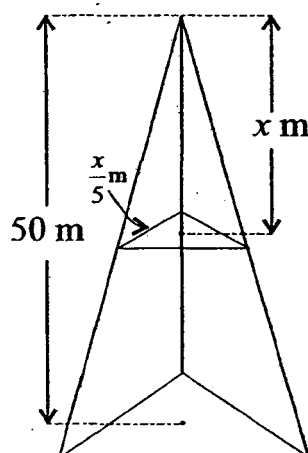
- (a) Consider the polynomial equation
- $x^3 - 3x^2 + x - 5 = 0$
- which has roots
- $\alpha$
- ,
- $\beta$
- and
- $\gamma$
- .

(i) Show that  $\alpha + \beta = 3 - \gamma$ .**1**

- (ii) Write down similar expressions for
- $\alpha + \gamma$
- and
- $\beta + \gamma$
- and hence find a polynomial equation which has the roots
- $\alpha + \beta$
- ,
- $\alpha + \gamma$
- and
- $\beta + \gamma$
- .

**2**

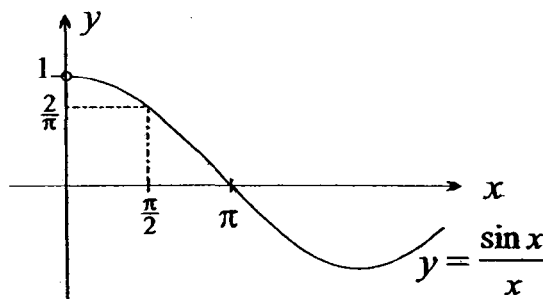
(b)



The diagram above shows a monument 50 metres high. A horizontal cross section  $x$  metres from the top is an equilateral triangle with sides  $\frac{x}{5}$  metres. Use integration to find the volume of the monument.

**3**

(c)



Use the method of cylindrical shells to find the volume of the solid formed when the region bounded by the curve  $y = \frac{\sin x}{x}$  and the lines  $y = 0$  and  $x = \frac{\pi}{2}$  is rotated about the  $y$ -axis.

**4**

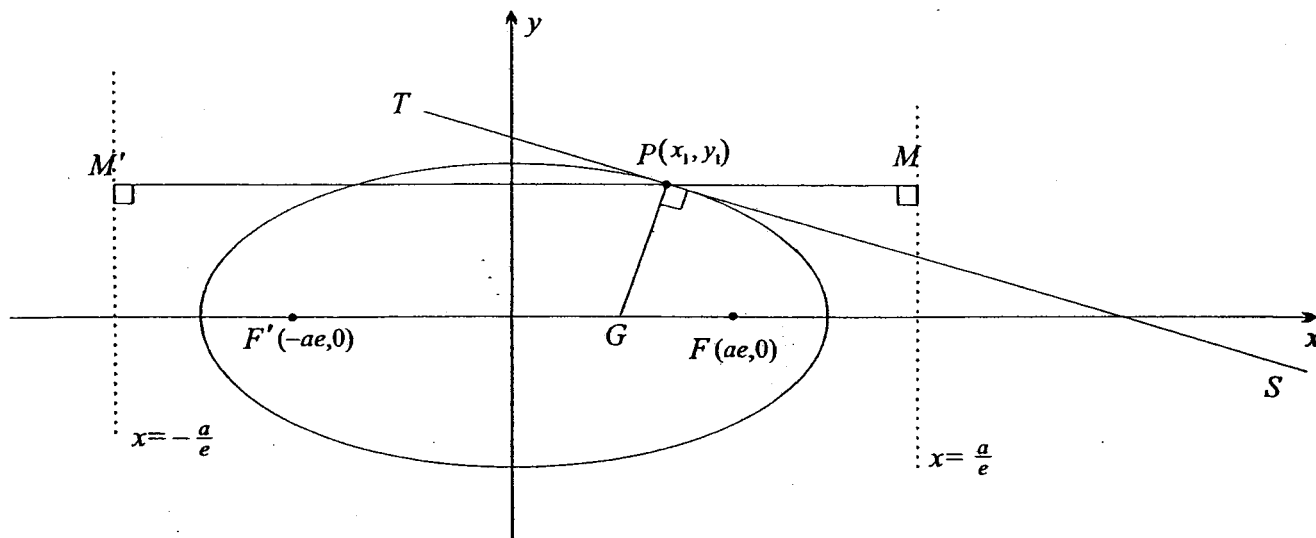
- (d) An hyperbola is defined parametrically by
- $x = 3 \sec \theta$
- and
- $y = 4 \tan \theta$
- .

(i) Write the equation of the curve in Cartesian form and show that the eccentricity is  $\frac{5}{3}$ .**2**(ii) Sketch the curve showing its  $x$ -intercepts, foci, directrices and asymptotes.**4**

Exam continues next page ...

**QUESTION FIVE** (15 marks) Use a separate writing booklet.**Marks**

(a)



The diagram above shows the ellipse  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$  with foci  $F(ae, 0)$  and  $F'(-ae, 0)$ .

$P(x_1, y_1)$  is any point on the ellipse.

Let  $M$  and  $M'$  be the feet of the perpendiculars from  $P$  to the directrices  $x = \frac{a}{e}$  and  $x = -\frac{a}{e}$ .

Line  $TS$  is a tangent to the ellipse at  $P$  and  $G$  is the point where the normal at  $P$  meets the  $x$ -axis.

(i) Show that the equation of the normal at  $P$  is  $\frac{a^2x}{x_1} - \frac{b^2y}{y_1} = a^2 - b^2$ .

3

(ii) Show that the point  $G$  has co-ordinates  $(e^2x_1, 0)$ .

3

(iii) Show that the distance  $PF$  is  $a - ex_1$ .

2

(iv) Show that  $\frac{PF}{FG} = \frac{PF'}{F'G}$ .

2

(b) (i) Show that  $1 - \cos 2\theta - i \sin 2\theta = 2 \sin \theta (\sin \theta - i \cos \theta)$ .

2

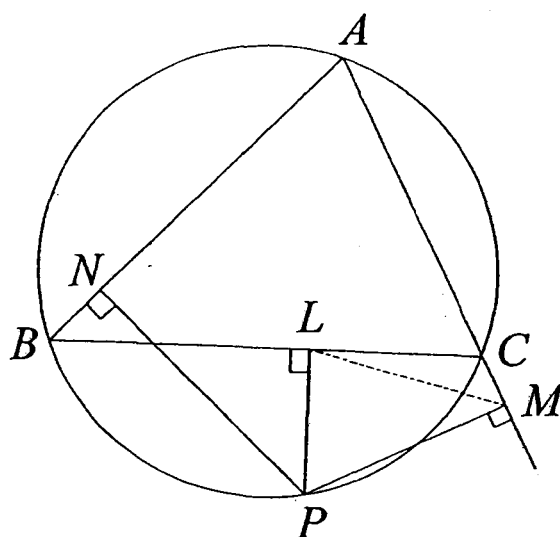
(ii) Given that  $\frac{z-1}{z} = \text{cis } \frac{2\pi}{5}$ , show that  $z = \frac{1}{2}(1 + i \cot \frac{\pi}{5})$ .

3

Exam continues overleaf ...

**QUESTION SIX** (15 marks) Use a separate writing booklet.**Marks**

(a)



In the diagram above,  $ABC$  is a triangle with the circumcircle through points  $A$ ,  $B$  and  $C$  drawn.  $P$  is another point on the minor arc  $BC$ . Points  $L$ ,  $M$  and  $N$  are the feet of the perpendiculars from  $P$  to the sides  $BC$ ,  $CA$  and  $AB$  respectively.

(i) Copy the diagram and explain why  $P$ ,  $L$ ,  $N$  and  $B$  are concyclic. 1

(ii) Explain why  $P$ ,  $L$ ,  $C$  and  $M$  are concyclic. 1

(iii) Let  $\angle PLM = \alpha$ .

( $\alpha$ ) Show that  $\angle ABP = \alpha$ . 2

( $\beta$ ) Hence show that  $M$ ,  $L$  and  $N$  are collinear. 2

- (b) A particle of unit mass is thrown vertically downwards with an initial velocity of  $v_0$ . It experiences a resistive force of magnitude  $kv^2$  where  $v$  is its velocity. Taking downwards as the positive direction, the equation of motion of the particle is given by

$$\ddot{x} = g - kv^2.$$

Let  $V$  be the terminal velocity of the particle.

(i) Explain why  $V = \sqrt{\frac{g}{k}}$ . 1

(ii) Show that  $v^2 = V^2 + (v_0^2 - V^2)e^{-2kx}$ . 4

- (c) Let  $z = x + iy$  be any non-zero complex number such that  $z + \frac{1}{z} = k$ , where  $k$  is a real number.

(i) Prove that either  $y = 0$  or  $x^2 + y^2 = 1$ . 2

(ii) Show that if  $y = 0$  then  $|k| \geq 2$ . 2

Exam continues next page ...

**QUESTION SEVEN** (15 marks) Use a separate writing booklet.**Marks**(a) (i) Write down  $\cos 2\theta$  in terms of  $\tan \theta$ .**1**(ii) Show that  $\cos 4\theta = \frac{1 - 6 \tan^2 \theta + \tan^4 \theta}{1 + 2 \tan^2 \theta + \tan^4 \theta}$ .**3**(iii) Deduce that  $\tan^2 \frac{\pi}{8} + \tan^2 \frac{3\pi}{8} = 6$ .**3**(b) Consider the equation  $z^7 = 1$ .This equation has seven roots  $1, \rho, \rho^2, \dots, \rho^6$ , where  $\rho = \text{cis } \frac{2\pi}{7}$ .Let  $\alpha = \rho + \rho^2 + \rho^4$  and  $\theta = \rho^3 + \rho^5 + \rho^6$ .(i) Express  $\rho^9$  as a lower positive power of  $\rho$ .**1**(ii) Simplify  $\alpha + \theta$ .**2**(iii) Simplify  $\alpha\theta$ .**2**(iv) Form a quadratic equation with  $\alpha$  and  $\theta$  as roots.**1**(v) Deduce that  $\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{8\pi}{7} = -\frac{1}{2}$ .**2****Exam continues overleaf ...**

**QUESTION EIGHT** (15 marks) Use a separate writing booklet.**Marks**

- (a) The sequence  $a_1, a_2, \dots, a_n$  is defined by  $a_n = \frac{(2n)!}{2^n n!}$ .

**4**

Show by induction on  $n$  that  $a_n$  is an odd positive integer.

- (b) Suppose that  $y = f(x)$  is an increasing function for  $x \geq 1$ .  
Suppose also that  $f(x) \geq 0$  for  $x \geq 1$ .

- (i) Explain, with the aid of a diagram, why

**2**

$$f(1) + f(2) + \dots + f(n-1) < \int_1^n f(x) dx < f(2) + f(3) + \dots + f(n).$$

- (ii) Show that  $\int_1^n \ln x dx = n \ln n - n + 1$ .

**2**

- (iii) Use parts (i) and (ii) to deduce that, for  $n > 1$ :

( $\alpha$ )  $n! > \frac{n^n}{e^{n-1}}$

**3**

( $\beta$ )  $n! < \frac{n^{n+1}}{e^{n-1}}$

**2**

- (iv) Find  $\lim_{n \rightarrow \infty} \frac{\sqrt[n]{n!}}{n}$ . (You may assume that  $\lim_{n \rightarrow \infty} \sqrt[n]{n} = 1$ .)

**2****END OF EXAMINATION**

## Question One

(1)

$$\begin{aligned}
 (a) \quad & \int_0^4 (2x+1)^{\frac{1}{2}} dx \\
 &= \left[ \frac{(2x+1)^{\frac{1}{2}}}{2 \times \frac{1}{2}} \right]_0^4 \quad \checkmark \\
 &= \sqrt{9} - \sqrt{1} \quad \checkmark \\
 &= 2.
 \end{aligned}$$

$$(b) \quad I = \int \tan^3 x \sec^2 x \, dx$$

$$\begin{aligned}
 \text{Let } u &= \tan x \\
 du &= \sec^2 x \, dx \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 I &= \int u^3 \, du \\
 &= \frac{u^4}{4} + C \\
 &= \frac{\tan^4 x}{4} + C \quad \checkmark
 \end{aligned}$$

$$(c) \quad \int \frac{x}{x^2 - 4x + 8} \, dx$$

$$= \frac{1}{2} \int \frac{2x - 4 + 4}{x^2 - 4x + 8} \, dx \quad \checkmark$$

$$= \frac{1}{2} \int \frac{2x - 4}{x^2 - 4x + 8} \, dx + \int \frac{2}{(x-2)^2 + 4} \, dx$$

$$= \frac{1}{2} \ln(x^2 - 4x + 8) + 2 \times \frac{1}{2} \tan^{-1} \frac{x-2}{2} + C$$

$$\begin{aligned}
 &= \frac{1}{2} \ln(x^2 - 4x + 8) + \tan^{-1} \frac{x-2}{2} + C \\
 &\quad \checkmark \quad \checkmark
 \end{aligned}$$

(2)

$$(d) \quad (1) \quad \frac{3x^2 - 10}{x^2 - 4x + 4} = 3 + \frac{A}{x-2} + \frac{B}{(x-2)^2}$$

$$3x^2 - 10 = 3(x-2)^2 + A(x-2) + B$$

$$\text{subst } x=2 : \quad 12 - 10 = B \quad \checkmark$$

$$B = 2$$

$$\text{subst } x=0 : \quad -10 = 12 + B - 2A$$

$$-22 = 2 - 2A \quad \checkmark$$

$$-24 = -2A \quad \checkmark$$

$$A = 12$$

$$\int \frac{3x^2 - 10}{x^2 - 4x + 4} dx = \int 3 + \frac{12}{x-2} + \frac{2}{(x-2)^2}$$

$$= 3x + 12 \ln|x-2| - \frac{4}{x-2} + C \quad \checkmark$$

$$(e) \quad \int_1^e \sin(\ln x) \frac{d}{dx}(x) dx$$

$$= x \sin(\ln x) \Big|_1^e - \int_1^e x \cdot \cos(\ln x) \cdot \frac{1}{x} dx \quad \checkmark$$

$$= x \sin(\ln x) \Big|_1^e - \int_1^e \cos \ln x \frac{d}{dx}(x) dx$$

$$= e \sin 1 - 0 - \left\{ [x \cos(\ln x)]_1^e - \int_1^e x \sin(\ln x) \cdot \frac{1}{x} dx \right\} \quad \checkmark$$

$$= e \sin 1 - \left\{ e \cos 1 - 1 + \int_1^e \sin(\ln x) dx \right\}$$

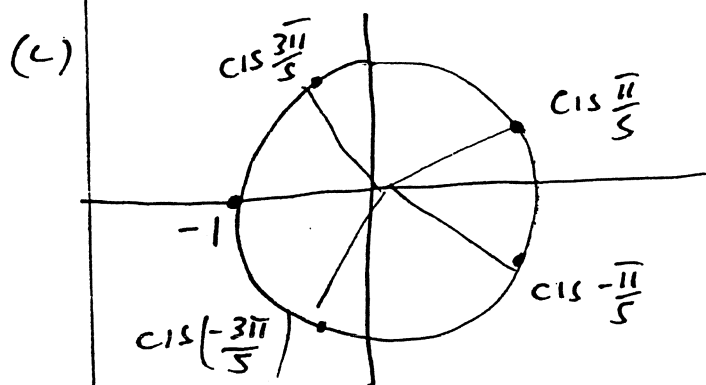
$$\text{So } 2 \int_1^e \sin(\ln x) dx = e \sin 1 - e \cos 1 + 1 \quad \checkmark$$

$$\int_1^e \sin(\ln x) dx = \frac{e}{2} [\sin 1 - \cos 1] + \frac{1}{2} \quad \checkmark$$

## Question Two

(a)  $|\cos \theta + i \sin \theta| = 1$  ✓

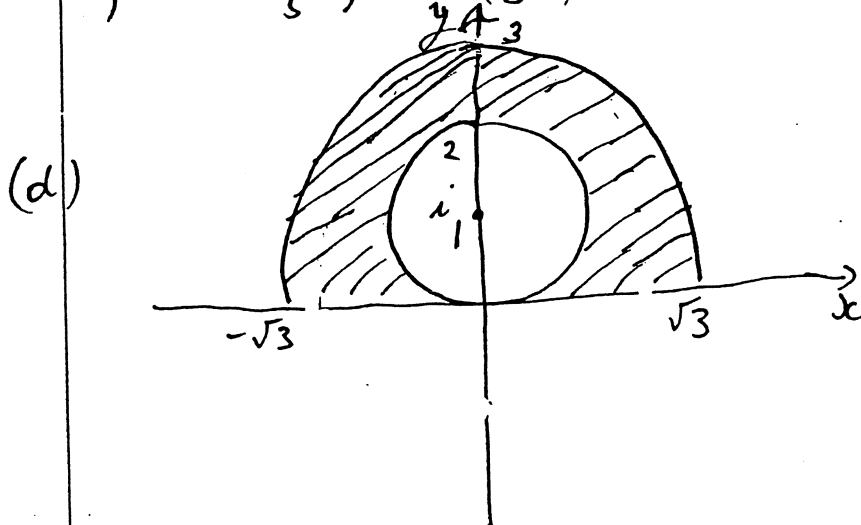
(b) 
$$\begin{aligned} \frac{z^5(1-z)}{2+i} &= \frac{z(1-z)}{2+i} \\ &= \frac{z+1}{2+i} \\ &= \frac{(1+i)(2-z)}{4+1} \quad \checkmark \\ &= \frac{2-z+2i+1}{5} \\ &= \frac{3}{5} + \frac{1}{5}i \quad \checkmark \end{aligned}$$



Clearly  $z = -1$  is a root.  
The other four roots are equally spaced around the circle.  
✓

Solutions to  $z^5 = -1$  are ✓

$-1, \text{cis } \frac{\pi}{5}, \text{cis}(-\frac{\pi}{5}), \text{cis } \frac{3\pi}{5} \text{ and } \text{cis}(-\frac{3\pi}{5})$



✓ annulus (0,1) centre  
✓ above x axis  
✓ correct shading (radius between 1 and 2)

(e)  $1+i$  is a root

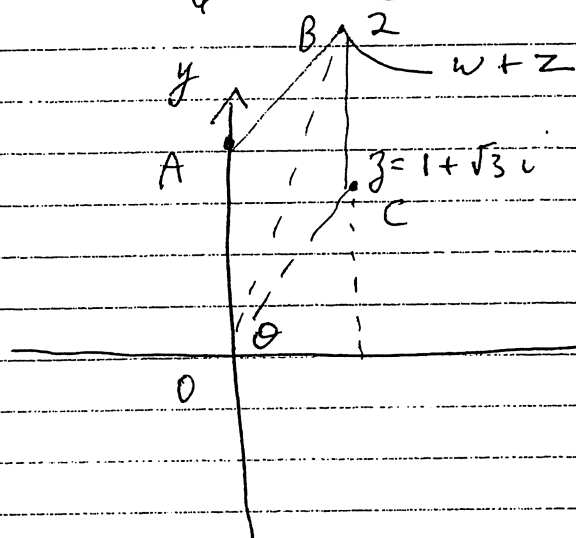
$$\therefore (1+i)^2 + \phi(1+i) - i = 0 \quad \checkmark$$

$$1+2i-1 + \phi(1+i) - i = 0$$

$$\phi = \frac{-i}{1+i} \times \frac{1-i}{1-i}$$

$$\phi = \frac{-1-i}{2} \quad \checkmark$$

(f)



(i)  $\tan \theta = \sqrt{3}$   
 $\theta = \frac{\pi}{3}$

$\arg(z) = \frac{\pi}{3} \quad \checkmark$

(ii) Multiplication by  $\text{cis } \alpha$  is a rotation anticlockwise through  $\alpha$ .  
 $\alpha + \frac{\pi}{3} = \frac{\pi}{2} \quad \checkmark$

So if  $w$  is purely imaginary and  $\text{Im}(z) > 0$   
 then  $\alpha = \frac{\pi}{2} - \frac{\pi}{3} = \frac{\pi}{6} \quad \checkmark$

(iii)  $OABC$  is a rhombus  
 Diagonals bisect vertex  $\checkmark$

So  $\arg(z+w) = \frac{\pi}{2} + \frac{\pi}{12}$   
 $= \frac{5\pi}{12} \quad \checkmark$

(5)

$$\begin{aligned}
 (a) \quad (i) \quad A(y) &= \pi(r_1^2 - r_2^2) \\
 &= \pi(3^2 - (2-y^2)^2) \quad \checkmark \\
 &= \pi(9 - 4 + 4y^2 - y^4) \\
 &= \pi(5 + 4y^2 - y^4) \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad V &= 2\pi \int_0^1 (5 + 4y^2 - y^4) dy \quad \checkmark \\
 &= 2\pi \left[ 5y + \frac{4y^3}{3} - \frac{y^5}{5} \right]_0^1 \\
 &= 2\pi \left[ 5 + \frac{4}{3} - \frac{1}{5} \right] \\
 &= \frac{2\pi [75 + 20 - 3]}{15} = \frac{184\pi}{15} \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad (i) \quad f(x) &= (1+x^3)^{-1} \\
 f'(x) &= -\frac{3x^2}{(1+x^3)^2} \quad \checkmark
 \end{aligned}$$

Start point where  $x = 0$

Horizontal point  
of inflection  
where  $x = 0$   
 $y = 1$

| $x$  | $-\frac{1}{2}$ | $0$ | $\frac{1}{2}$ |
|------|----------------|-----|---------------|
| $y'$ | -ve            | 0   | -ve           |

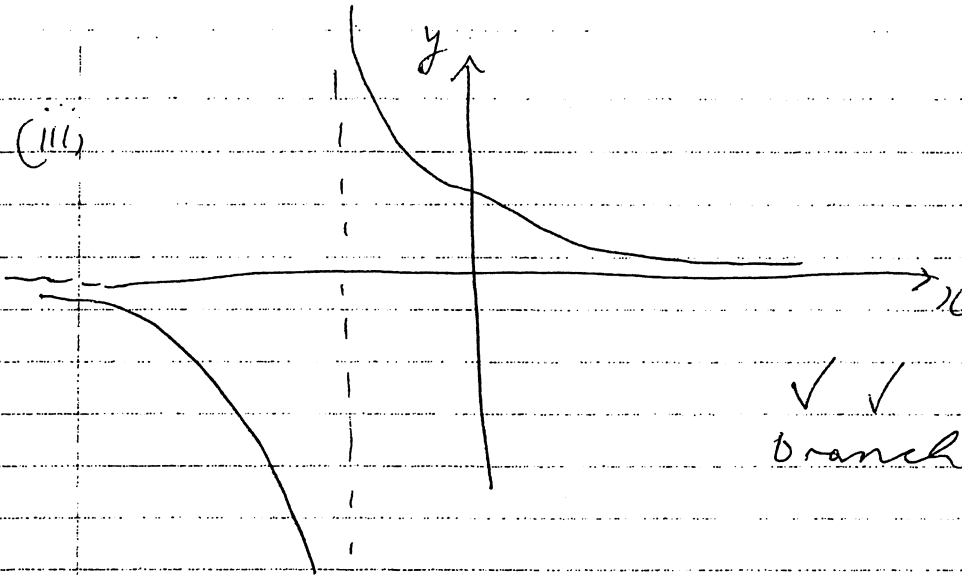
(ii) Vertical asymptote at  $x = -1$   $\checkmark$

As  $x \rightarrow \infty$ ,  $y \rightarrow 0^+$   
 $x \rightarrow -\infty$ ,  $y \rightarrow 0^-$   $\checkmark$

Horizontal asymptote at  $y = 0$ .

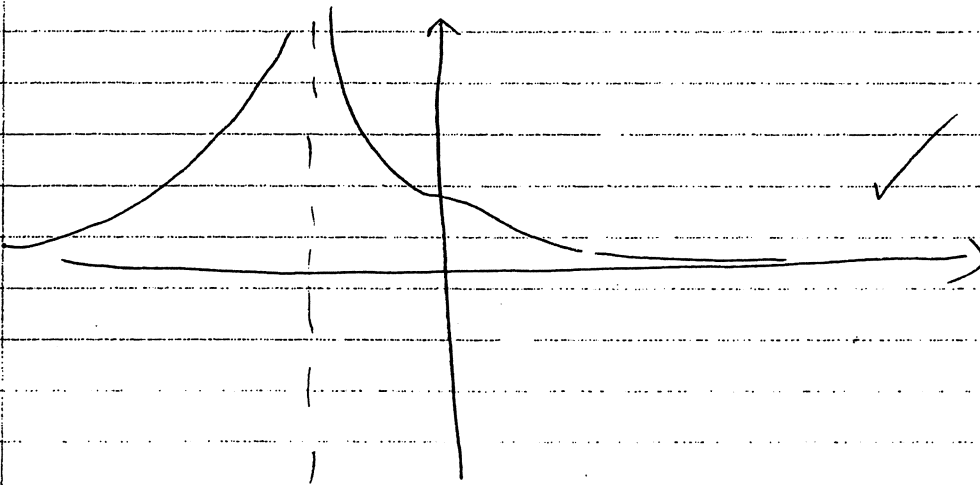
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(iii)



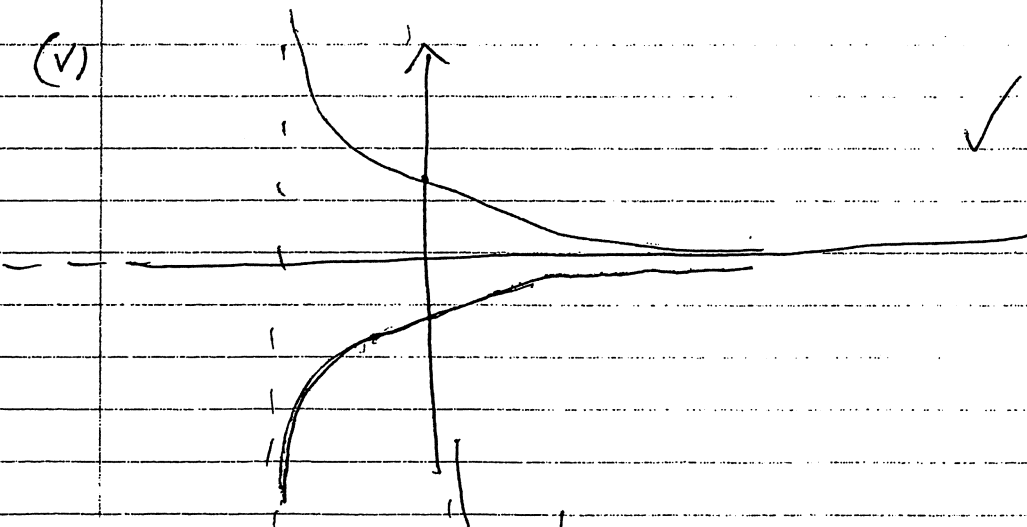
✓ ✓ for each  
branch correct.

(iv)



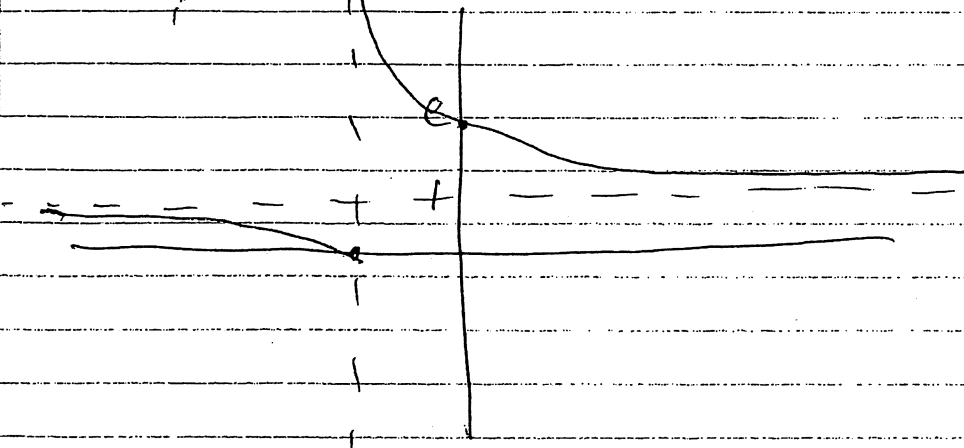
✓

(v)



correct domain  
✓ ✓ reflection.

(vi)



✓ ✓

(7)

Question Four

(a)  $x^3 - 3x^2 + x - 5 = 0$

(i)  $\alpha + \beta + \gamma = -\frac{b}{a}$

$\alpha + \beta + \gamma = 3$  ✓

$\alpha + \beta = 3 - \gamma$

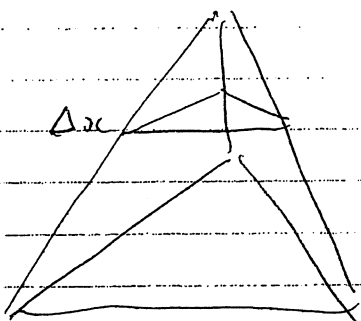
(ii)  $\beta + \gamma = 3 - \alpha$  and  $\alpha + \gamma = 3 - \beta$

The polynomial equation has roots  
 $3 - \alpha$ ,  $3 - \beta$ ,  $3 - \gamma$ .

Transformation is  $y = 3 - x$   
 $x = 3 - y$  ✓

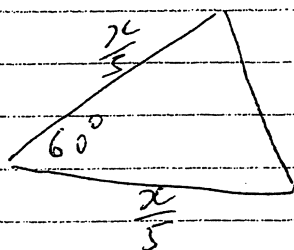
New equation is  $(3 - y)^3 - 3(3 - y)^2 + (3 - y) - 5 = 0$  ✓

(b)



$V = \lim_{\Delta x \rightarrow 0} \sum A(x) \Delta x$

$V = \int_0^{50} \frac{1}{2} \left( \frac{x}{5} \right)^2 \cdot \frac{\sqrt{3}}{2} dx$  ✓



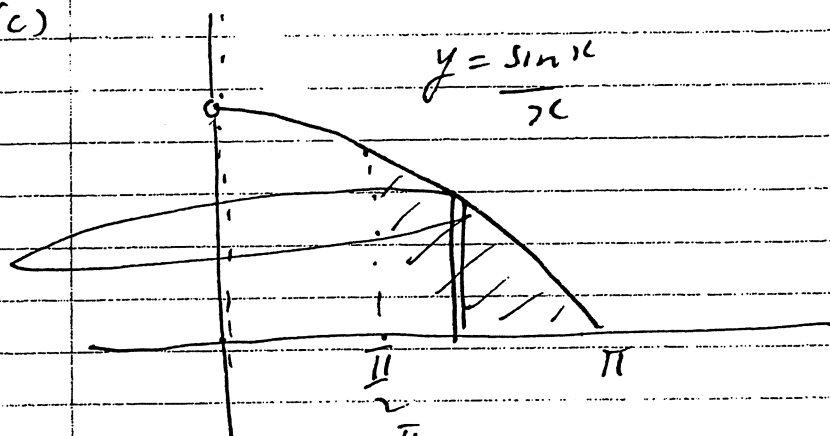
$A(x) = \frac{1}{2} \left( \frac{x}{5} \right)^2 \sin 60^\circ$   
 $= \frac{x^2}{50} \cdot \frac{\sqrt{3}}{2}$  ✓

$= \frac{\sqrt{3}}{100} \left[ \frac{x^3}{3} \right]_0^{50}$

$= \frac{\sqrt{3}}{100} \times \frac{50^3}{3}$

$= \frac{1250\sqrt{3}}{6} \text{ units}^3$  ✓

(c)



$$V = 2\pi \int_{\frac{\pi}{2}}^{\pi} x \cdot \frac{\sin x}{x} dx$$

$$= 2\pi \int_{\frac{\pi}{2}}^{\pi} \sin x dx$$

$$= 2\pi \left[ -\cos x \right]_{\frac{\pi}{2}}^{\pi}$$

$$= 2\pi [-1 - 0] = 2\pi$$

(d)

$$x = 3 \sec \theta \quad y = 4 \tan \theta$$

(i)

$$\frac{x^2}{9} - \frac{y^2}{1} = 1$$

$$a^2 = 9, \quad b^2 = 1$$

$$b^2 = a^2(e^2 - 1)$$

$$16 = 9(e^2 - 1)$$

$$\frac{16}{9} + 1 = e^2$$

$$e^2 = \frac{25}{9}$$

$$e = \frac{5}{3}$$

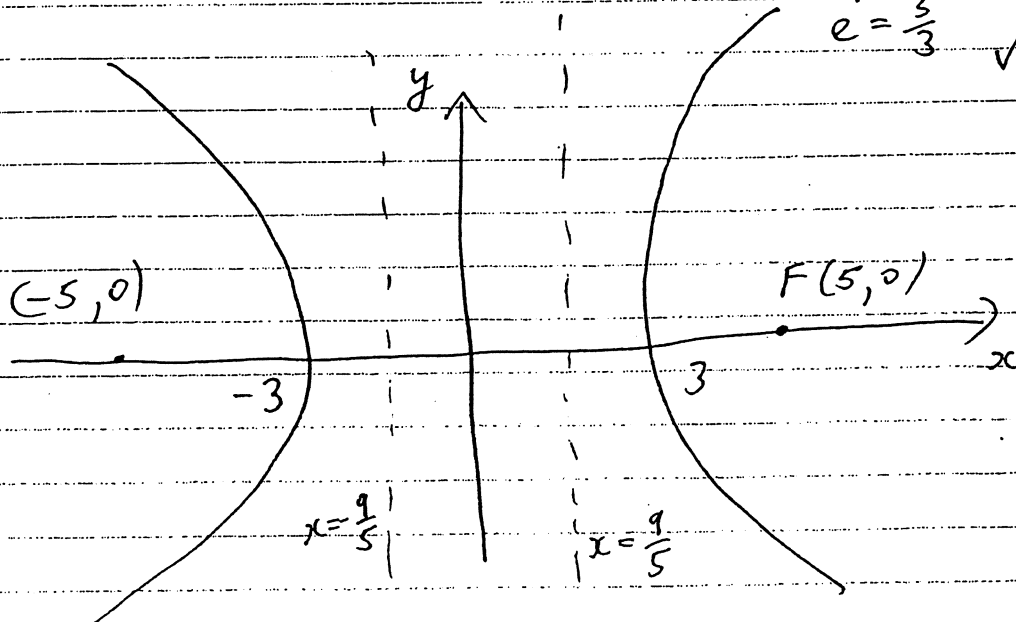
(ii)

x-ints ✓

foci ✓

directrices ✓

asymptotes ✓



Question Five

(a)  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

(i) Differentiate implicitly wrt  $x$

$$\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{b^2 x}{a^2 y}$$

At  $P(x_1, y_1)$ , gradient  $= -\frac{b^2 x_1}{a^2 y_1}$

So gradient of normal is  $\frac{a^2 y_1}{b^2 x_1}$

Eqn of normal is

$$y - y_1 = \frac{a^2 y_1}{b^2 x_1} (x - x_1)$$

$$\frac{y}{y_1} - 1 = \frac{a^2}{b^2} \left( \frac{x}{x_1} - 1 \right)$$

$$\frac{a^2 x}{x_1} - \frac{b^2 y}{y_1} = a^2 - b^2$$

(ii) When  $y = 0$ ,  $x = \frac{a^2 - b^2}{a^2} x_1$

But  $\frac{a^2 - b^2}{a^2} = e^2$

So  $G = (e^2 x_1, 0)$

(iii) Now  $PF = e \cdot PM$  (defn of ellipse)

$$PF = e \left( \frac{a}{e} - x_1 \right)$$

$$= a - e x_1$$

$$(iv) \quad \frac{PF}{FG} = \frac{a - ex_1}{ae - e^2 x_1} = \frac{1}{e} \quad \checkmark$$

$$\frac{PF'}{F'G} = \frac{a + ex_1}{a + e^2 x_1} = \frac{1}{e} \quad \checkmark$$

$$\text{So } \frac{PF}{FG} = \frac{PF'}{F'G}$$

$$\begin{aligned} (b) \quad (i) \quad LHS &= 1 - \cos 2\alpha - i \sin 2\alpha \\ &= 1 - (1 - 2\sin^2 \alpha) - 2i \sin \alpha \cos \alpha \quad \checkmark \\ &= 2\sin^2 \alpha - 2i \sin \alpha \cos \alpha \\ &= 2\sin \alpha (\sin \alpha - i \cos \alpha) \quad \checkmark \\ &= RHS \end{aligned}$$

$$(ii) \quad \frac{z-1}{z} = \operatorname{cis} \frac{2\pi}{5}$$

$$z - 1 = z \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5}$$

$$z (1 - \cos \frac{2\pi}{5} - i \sin \frac{2\pi}{5}) = 1$$

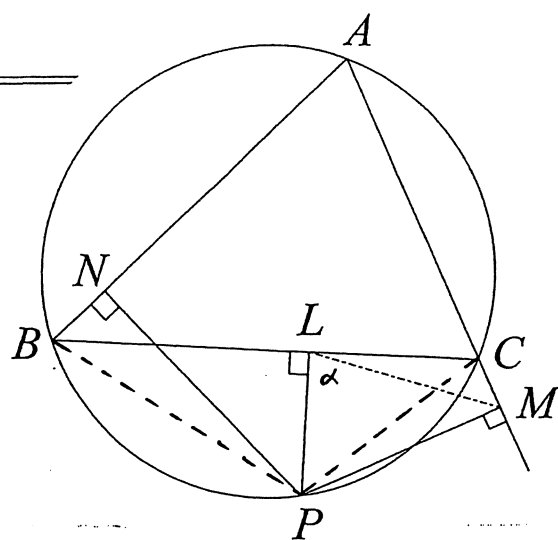
$$z \times 2 \sin \frac{\pi}{5} (\sin \frac{\pi}{5} - i \cos \frac{\pi}{5}) = 1 \quad \text{by part (i)} \quad \checkmark$$

$$2z \sin \frac{\pi}{5} = \sin \frac{\pi}{5} + i \cos \frac{\pi}{5} \quad \checkmark$$

$$z = \frac{1}{2} (1 + i \cot \frac{\pi}{5}) \quad \checkmark$$

Q 6

a)



(i)  $\angle BLP = \angle BNP$  (given) ✓

So B, L, N and P are concyclic by converse of angles standing on the same arc

(ii)  $\angle PLC + \angle PMC = 180^\circ$  (given)

So P, L, C and M are concyclic by converse of opposite angles of a cyclic quad ✓

(iii) (a)  $\angle PCM = \angle PLM$  (angles standing on the same arc PM) ✓  
 $= \alpha$

$\angle ABP = \angle PCM$  (opposite interior angle of cyclic quad ABPC) ✓  
 $= \alpha$

(b)  $\angle NBP = \alpha$  (same as  $\angle ABP$ )

So  $\angle NLP = 180^\circ - \alpha$  (opp angles of cyclic quad BNLP) ✓

So  $\angle NLM = \angle NLP + \angle MLP$   
 $= 180^\circ - \alpha + \alpha$   
 $= 180^\circ$  ✓

$\therefore$  N, L and M are collinear.

b) (i) Terminal velocity when  $ac = 0$

$g - KV^2 = 0$  ✓

$V^2 = \frac{g}{K}$

So  $V = \sqrt{\frac{g}{K}}$  (since  $V_0 > 0$  initially and so cannot change sign)

(b) (ii)  $v \frac{dv}{dx} = g - kv^2$

$$\frac{dx}{dv} = \frac{v}{g - kv^2}$$

$$\frac{dx}{dv} = -\frac{1}{2k} \frac{-2kv}{g - kv^2}$$

$$x = -\frac{1}{2k} (g - kv^2) + c$$

When  $x=0$  }  $c = \frac{1}{2k} (g - kv_0^2)$   
 $v=v_0$

$$x = -\frac{1}{2k} \ln \left( \frac{g - kv^2}{g - kv_0^2} \right)$$

$$\frac{g - kv^2}{g - kv_0^2} = e^{-2kx}$$

$$g - kv^2 = (g - kv_0^2) e^{-2kx}$$

$$-kv^2 = -g + (g - kv_0^2) e^{-2kx}$$

$$v^2 = \frac{g}{k} + \left( v_0^2 - \frac{g}{k} \right) e^{-2kx}$$

$$v^2 = v_0^2 + (v_0^2 - v^2) e^{-2kx}$$

(c) (i) Let  $z = x + iy$

$$z = x + iy + \frac{1}{x + iy} \times \frac{x - iy}{x^2 - iy^2}$$

$\text{Im}(z) = 0$  so  $y + \frac{-y}{x^2 + y^2} = 0$

$$y \left( 1 - \frac{1}{x^2 + y^2} \right) = 0$$

$$y = 0 \text{ or } x^2 + y^2 = 1$$

(ii)  $x + \frac{1}{x} = k$

$$x^2 - kx + 1 = 0$$

has real roots.

$$\therefore \Delta \geq 0$$

$$k^2 - 4 \geq 0$$

$$|k| \geq 2$$

Question Seven

$$(a) \quad (i) \quad \cos 2\theta = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} \quad \checkmark$$

$$(ii) \quad \cos 4\theta = \frac{1 - \tan^2 2\theta}{1 + \tan^2 2\theta} \quad \checkmark$$

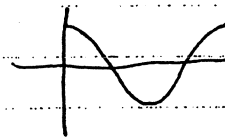
$$= \frac{1 - \left( \frac{2 \tan \theta}{1 - \tan^2 \theta} \right)^2}{1 + \left( \frac{2 \tan \theta}{1 - \tan^2 \theta} \right)^2} \quad \checkmark$$

$$= \frac{1 - 2 \tan^2 \theta + \tan^4 \theta - 4 \tan^2 \theta}{1 - 2 \tan^2 \theta + \tan^4 \theta + 4 \tan^2 \theta}$$

$$= \frac{1 - 6 \tan^2 \theta + \tan^4 \theta}{1 + 2 \tan^2 \theta + \tan^4 \theta} \quad \checkmark$$

Consider  $\cos 4\theta = 0$   
 $4\theta = \frac{\pi}{2} \text{ or } \frac{3\pi}{2}$

$$\theta = \frac{\pi}{8}, \frac{3\pi}{8}$$



Consider the quadratic

$$1 - 6x + x^2 = 0 \quad \text{---} \quad *$$

where  $x = \tan^2 \theta$

The solutions to  $*$  are

$$\tan^2 \frac{\pi}{8} \quad \text{and} \quad \tan^2 \frac{3\pi}{8}$$

$$\Sigma \text{ roots} = \tan^2 \frac{\pi}{8} + \tan^2 \frac{3\pi}{8} = -\frac{b}{a}$$

$$\tan^2 \frac{\pi}{8} + \tan^2 \frac{3\pi}{8} = 6 \quad \checkmark$$

# Question Seven (b)

14

$$(i) \quad \rho^9 = \rho^7 \times \rho^2 \\ = \rho^2 \quad \checkmark$$

$$(ii) \quad \alpha + \theta = \rho + \rho^2 + \rho^3 + \rho^4 + \rho^5 + \rho^6$$

In the equation  $z^7 - 1 = 0$   $\checkmark$   
coeff of  $x^6 = 0$

$$\therefore 1 + \rho + \rho^2 + \rho^3 + \rho^4 + \rho^5 + \rho^6 = 0$$

$$\rho + \rho^2 + \rho^3 + \rho^4 + \rho^5 + \rho^6 = -1 \quad \checkmark$$

$$(iii) \quad \alpha\theta = (\rho + \rho^2 + \rho^4)(\rho^3 + \rho^5 + \rho^6)$$

$$= \rho^4 + \rho^6 + \rho^7 + \rho^5 + \rho^7 + \rho^8 + \rho^7 + \rho^9 + \rho^{10}$$

$$= \rho + \rho^2 + \rho^3 + \rho^4 + \rho^5 + \rho^6 + \rho^7 + 2 \quad \checkmark$$

$$= 0 + 2 \quad \checkmark$$

$$= 2$$

(iv) The quadratic is

$$x^2 - (\alpha + \theta)x + \alpha\theta = 0 \quad \checkmark$$

$$x^2 + x + 2 = 0 \quad \text{---} *$$

(v) The roots of \* are

$$x = \frac{-1 \pm \sqrt{7}i}{2} \quad \checkmark$$

$$\operatorname{Re}(\rho + \rho^2 + \rho^4) = -\frac{1}{2}$$

$$\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{8\pi}{7} = -\frac{1}{2} \quad \checkmark$$

3 (a)  $a_1 = \frac{2!}{2^1 \times 1!} = 1$ , which is odd. ✓

Suppose that  $a_k = \frac{(2k)!}{2^k k!}$  is odd. ✓

Prove that  $a_{k+1}$  is odd:

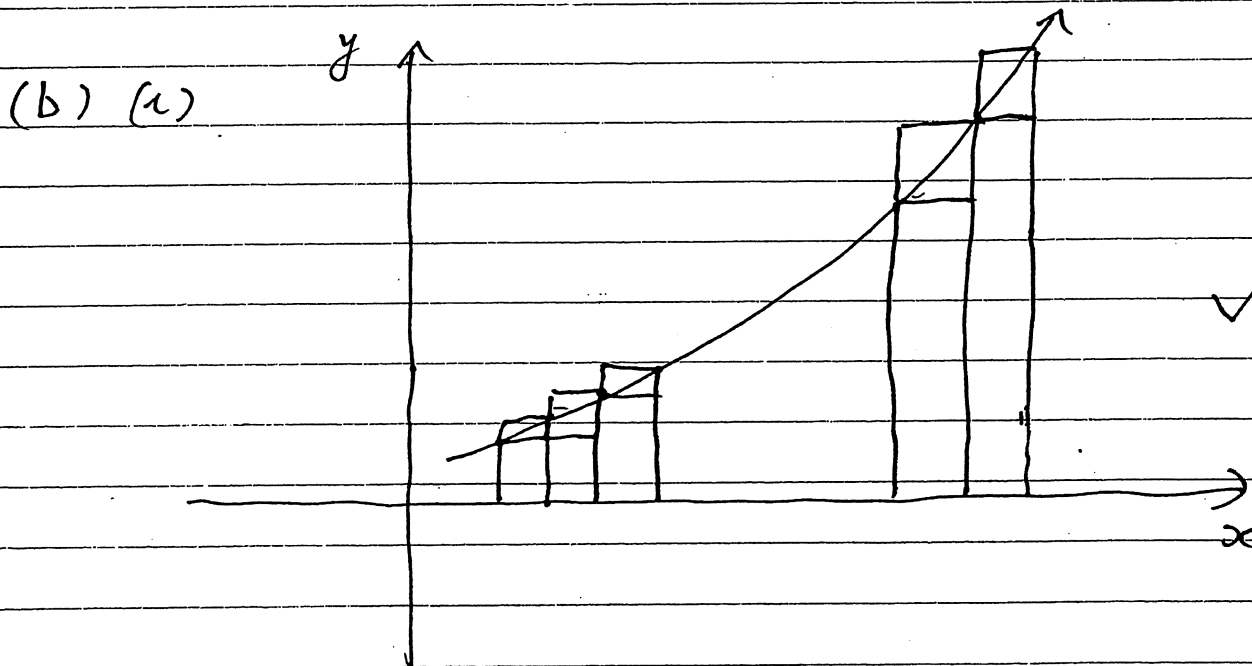
$$a_{k+1} = \frac{(2k+2)!}{2^{k+1} (k+1)!}$$

$$= \frac{(2k+2)(2k+1)(2k)!}{2 \times 2^k \times (k+1) \times k!}$$

$$= \frac{2(k+1)(2k+1)(2k)!}{2(k+1) 2^k k!}$$

$$= \underbrace{(2k+1)}_{\text{odd}} \cdot \underbrace{a_k}_{\text{odd}}$$

$\therefore a_{k+1}$  is odd ✓



sum of area  
of lower rectangles

< exact area  
under curve  
from  $x=1$  to  $x=n$

< sum of areas  
of upper rectangles ✓

$$f(1) + f(2) + \dots + f(n-1) < \int_1^n f(x) dx < f(2) + f(3) + \dots + f(n)$$

$$(ii) \int_1^n \ln x \frac{d}{dx}(x) dx$$

$$= [x \ln x]_1^n - \int_1^n 1 dx \quad \checkmark$$

$$= n \ln n - n + 1 \quad \checkmark$$

$$(iii) (x) \text{ Let } f(x) = \ln x$$

$$\text{From (i): } \int_1^n \ln x dx < \ln 2 + \ln 3 + \dots + \ln n$$

$$\text{Using (ii): } n \ln n - n + 1 < \ln n! \quad \checkmark$$

$$n! > e^{n \ln n - n + 1}$$

$$n! > n^n e^{1-n} \quad \checkmark$$

$$n! > \frac{n^n}{e^{n-1}} \quad \checkmark$$

$$(\beta) \text{ From (i) and (ii)}$$

$$\ln 1 + \ln 2 + \dots + \ln(n-1) < n \ln n - n + 1 \quad \checkmark$$

$$\ln(n-1)! < n \ln n - n + 1$$

$$(n-1)! < n^n \cdot e^{1-n}$$

$$n! < \frac{n^{n+1}}{e^{n-1}} \quad \checkmark$$

$$(iv) \text{ From (iii), } \frac{n^n}{e^{n-1}} < n! < \frac{n^{n+1}}{e^{n-1}}$$

$$\text{Taking } n^{\text{th}} \text{ roots: } \frac{n}{e^{1-\frac{1}{n}}} < (n!)^{\frac{1}{n}} < \frac{n^{1+\frac{1}{n}}}{e^{1-\frac{1}{n}}} \quad \checkmark$$

$$\therefore e^{\frac{1}{n}-1} < \frac{n}{\sqrt[n]{n!}} < \sqrt[n]{n} e^{\frac{1}{n}-1}$$

$$\text{As } n \rightarrow \infty \quad e^{\frac{1}{n}-1} \rightarrow e^{-1} \text{ and } \sqrt[n]{n} \rightarrow 1 \text{ given}$$

$$\therefore \lim_{n \rightarrow \infty} \frac{\sqrt[n]{n!}}{n} = \frac{1}{e} \quad \checkmark$$